200 Most Important Physics MCQs — TU IOST BSc CSIT Entrance 2083

(High-Yield Master Set — Weighted by PYQ Frequency 2069–2081)

Distribution based on topic weightage analysis: Heat/Thermo (18), Current Electricity (16), Electrostatics (14), Ray Optics (14), Magnetism/EMI (14), Modern Physics (16), Nuclear Physics (10), Mechanics/Rotation (16), Kinematics/Newton's Laws (14), Circular Motion/Gravitation (12), SHM/Waves/Sound (16), Fluid Mechanics/Elasticity (10), Units/Dimensions (8), AC Circuits (6), Semiconductors (6).

1. Units, Dimensions & Errors (Q1–8)

1. Dimension of angular momentum is a) $[ML^2T^{-1}]$ b) $[MLT^{-1}]$ c) $[ML^2T^{-2}]$ d) $[MLT^{-2}]$
2. Which pair has the same dimensions as work? a) Torque b) Power c) Momentum d) Pressure
3. Number of significant figures in 0.0060 is a) 2 b) 3 c) 4 d) 5
4. Dimension of Planck's constant is same as that of a) Energy b) Angular momentum c) Force d) Power
5. Percentage error in kinetic energy for 2% error in velocity is a) 2% b) 4% c) 1% d) 8%
6. The dimensional formula of surface tension is a) $[MT^{-2}]$ b) $[ML^{-1}T^{-2}]$ c) $[MLT^{-2}]$ d) $[ML^2T^{-2}]$
7. Dimension of coefficient of viscosity is a) $[ML^{-1}T^{-1}]$ b) $[MLT^{-1}]$ c) $[ML^{-1}T^{-2}]$ d) $[ML^2T^{-1}]$
8. Unit of universal gravitational constant G is a) Nm^2/kg^2 b) Nm/kg c) N/kg^2 d) Nm^2/kg

2. Kinematics & Newton's Laws (Q9–22)

9. A stone dropped from height h reaches ground with speed v. From height 4h, speed will be a) 2v b) 4v c) v/2 d) 16v
10. Maximum range of a projectile is at angle a) 30° b) 45° c) 60° d) 90°
11. A body starts from rest with uniform acceleration a. Distance in nth second is proportional to a) n b) n^2 c) (2n−1) d) 1/n
12. Newton's second law states F = a) ma b) m/a c) m^2a d) ma^2
13. A lift accelerates upward with acceleration a. Apparent weight of person of mass m is a) m(g−a) b) m(g+a) c) mg d) ma
14. If a lift falls freely, apparent weight is a) mg b) zero c) 2mg d) mg/2
15. Rocket propulsion is based on conservation of a) energy b) momentum c) mass d) charge
16. Two masses m_1, m_2 connected via frictionless pulley. Acceleration of system is a) $(m_1-m_2)g/(m_1+m_2)$ b) $(m_1+m_2)g$ c) m_1g/m_2 d) g
17. Coefficient of static friction is generally a) less than kinetic friction b) ≥ kinetic friction c) zero d) equal to normal force
18. A bullet embeds in a stationary block (perfectly inelastic collision). Common velocity is given by a) mv/(m+M) b) Mv/(m+M) c) mv/M d) v
19. Impulse equals a) change in momentum b) force × velocity c) mass × displacement d) energy/time

20. A body moving with constant velocity has a) zero net force b) nonzero net force c) increasing speed d) variable direction
21. Angle of friction θ relates to coefficient of friction μ by a) $\tan\theta = \mu$ b) $\sin\theta = \mu$ c) $\cos\theta = \mu$ d) $\cot\theta = \mu$
22. For a projectile, time of flight is given by a) $2u \sin\theta/g$ b) $u^2\sin2\theta/g$ c) $u \sin\theta/g$ d) $2u \cos\theta/g$

3. Circular Motion & Gravitation (Q23–34)

23. Centripetal acceleration is directed a) along velocity b) towards center c) away from center d) tangentially
24. Orbital velocity of a satellite near earth's surface is a) $\sqrt{gR}$ b) $\sqrt{2gR}$ c) gR d) $\sqrt{gR}/2$
25. Escape velocity relates to orbital velocity (near surface) by a) $v_e = \sqrt{2}\, v_o$ b) $v_e = v_o$ c) $v_e = v_o/\sqrt{2}$ d) $v_e = 2v_o$
26. Kepler's third law states $T^2 \propto$ a) r b) r^2 c) r^3 d) $1/r^2$
27. Weightlessness in a satellite occurs because a) gravity is zero there b) gravity provides exactly the centripetal force c) no air resistance d) high speed cancels gravity
28. Value of g at the poles compared to the equator is a) less b) more c) equal d) zero
29. If earth's radius decreases by 1% (mass constant), g a) increases by 2% b) decreases by 2% c) increases by 1% d) remains same
30. A geostationary satellite has time period a) 24 hours b) 12 hours c) 365 days d) 1 hour
31. A car takes a banked turn without friction; banking angle θ satisfies a) $\tan\theta = v^2/rg$ b) $\tan\theta = rg/v^2$ c) $\sin\theta = v^2/rg$ d) $\cos\theta = v^2/rg$
32. Two satellites orbit at radii r and 4r. Ratio of orbital speeds is a) 2:1 b) 1:2 c) 4:1 d) 1:4
33. Gravitational PE of a body at height h ($h \ll R$) is a) mgh b) $-GMm/R$ c) $-GMm/(R+h)$ d) GMm/R^2
34. If a body is taken to a height equal to earth's radius, weight becomes a) mg/2 b) mg/4 c) mg d) 2mg

4. Rotational Mechanics (Q35–48)

35. Moment of inertia of a solid sphere about its diameter is a) $(2/5)MR^2$ b) $(2/3)MR^2$ c) MR^2 d) $(7/5)MR^2$
36. Moment of inertia of a ring about its axis is a) MR^2 b) $(1/2)MR^2$ c) $(2/5)MR^2$ d) $(1/4)MR^2$
37. Relation between torque τ, moment of inertia I, angular acceleration α is a) $\tau = I\alpha$ b) $\tau = I/\alpha$ c) $\tau = I\omega$ d) $\tau = I^2\alpha$
38. Angular momentum is conserved when a) net force is zero b) net torque is zero c) mass is constant d) velocity is constant
39. A solid cylinder rolls down an incline without slipping. Its acceleration is a) $g \sin\theta$ b) $(2/3)g \sin\theta$ c) g d) $(1/2)g \sin\theta$
40. A constant torque changes angular momentum from L_o to $4L_o$ in 4 sec. Torque is a) L_o b) $(3/4)L_o$ c) $4L_o$ d) $12L_o$

41. Kinetic energy of a rolling solid sphere (translation+rotation) is a) $(7/10)Mv^2$ b) $(1/2)Mv^2$ c) Mv^2 d) $(2/5)Mv^2$
42. A ballet dancer spins faster when she a) stretches arms out b) pulls arms in c) stops moving d) jumps
43. The parallel axis theorem states $I = I_{cm}$ + a) Md^2 b) Md c) M^2d^2 d) $Md^2/2$
44. Two discs of same mass, radii R and 2R rotate with same ω. Ratio of KE is a) 1:2 b) 1:4 c) 2:1 d) 4:1
45. Moment of inertia of a uniform rod (length L) about one end is a) $ML^2/12$ b) $ML^2/3$ c) ML^2 d) $ML^2/4$
46. A rigid body is in rotational equilibrium when a) net force is zero b) net torque is zero c) both a and b d) neither
47. Radius of gyration depends on a) mass only b) distribution of mass and axis c) angular velocity d) torque
48. Perpendicular axis theorem applies to a) 3D bodies only b) planar (2D) bodies c) rings only d) spheres only

5. Work, Energy, Power & Elasticity (Q49–62)

49. Work-energy theorem states work done equals a) change in momentum b) change in kinetic energy c) change in PE d) power × time
50. Kinetic energy of a body increases 4 times. Momentum increases by a) 2 times b) 4 times c) 16 times d) $\sqrt{2}$ times
51. In a perfectly inelastic collision, a) KE is conserved b) momentum conserved, KE not conserved c) both conserved d) neither conserved
52. Coefficient of restitution for perfectly elastic collision is a) 0 b) 1 c) between 0 and 1 d) infinite
53. SI unit of power is a) Joule b) Watt c) Newton d) Joule-second
54. 1 horsepower is approximately a) 550 W b) 746 W c) 1000 W d) 100 W
55. Young's modulus is the ratio of a) shear stress to shear strain b) longitudinal stress to longitudinal strain c) volumetric stress to strain d) force to area
56. Bulk modulus of a perfectly rigid body is a) zero b) infinite c) unity d) negative
57. Poisson's ratio theoretically lies between a) 0 and 1 b) −1 and 0.5 c) 0 and 0.5 only d) −1 and 1
58. Elastic PE per unit volume of stretched wire is a) stress×strain b) (1/2)×stress×strain c) stress/strain d) 2×stress×strain
59. A wire is stretched within elastic limit. If length is doubled (same diameter), Young's modulus a) doubles b) halves c) remains same d) becomes 4 times
60. Potential energy of a spring stretched by x is a) kx b) $kx^2/2$ c) kx^2 d) $2kx^2$
61. A ball dropped from h bounces to h/2. Coefficient of restitution is a) 1/2 b) $1/\sqrt{2}$ c) 1/4 d) $\sqrt{2}$
62. Two bodies of mass m and 4m have equal KE. Ratio of momenta is a) 1:2 b) 1:4 c) 2:1 d) 4:1

6. Fluid Mechanics (Q63–72)

63. Bernoulli's theorem is based on conservation of a) mass b) momentum c) energy d) charge
64. Excess pressure inside a soap bubble of radius r is a) 2T/r b) 4T/r c) T/r d) T/2r
65. Viscosity of liquids with increase in temperature a) increases b) decreases c) remains same d) becomes infinite
66. Terminal velocity of a spherical body in viscous medium is proportional to a) r b) r^2 c) 1/r d) $1/r^2$
67. A body floats with 3/4 volume submerged in water. Its density is a) 0.25 g/cc b) 0.75 g/cc c) 1 g/cc d) 0.5 g/cc
68. Equation of continuity states for incompressible flow a) $A_1v_1 = A_2v_2$ b) $A_1/v_1 = A_2/v_2$ c) $A_1+v_1 = A_2+v_2$ d) $v_1 = v_2$
69. Capillary rise is inversely proportional to a) surface tension b) radius of tube c) density only d) atmospheric pressure
70. Upthrust on a body depends on a) shape of body b) volume of liquid displaced c) mass of body d) depth of immersion
71. Reynold's number is used to determine a) viscosity b) laminar or turbulent flow c) surface tension d) elasticity
72. Hydraulic lift works on the principle of a) Bernoulli's theorem b) Pascal's law c) Archimedes' principle d) Stokes' law

7. Heat & Thermodynamics — HIGHEST WEIGHTAGE (Q73–100)

73. A temperature of 68°F equals a) 20°C b) 15°C c) 25°C d) 30°C
74. Celsius and Fahrenheit scales show same reading at a) 0° b) 100° c) −40° d) 40°
75. Relation between α, β, γ (linear, areal, cubical expansivity) is a) β=2α, γ=3α b) β=3α, γ=2α c) α=β=γ d) β=α/2
76. An iron ball is heated. Percentage increase is largest in a) diameter b) surface area c) volume d) density
77. Material with lowest linear expansivity is a) aluminum b) invar c) steel d) copper
78. Specific heat capacity is heat required to raise temperature of a) 1 kg by 1°C b) unit mass by unit temperature c) both a and b d) 1 g by 1°C only
79. Principle of calorimetry states a) heat lost = heat gained b) heat lost > heat gained c) total heat increases d) heat is destroyed
80. 50g ice at 0°C mixed with 50g water at 90°C. Final temperature (latent heat=80cal/g) is approximately a) 0°C b) 15°C c) 25°C d) 50°C
81. Latent heat of fusion of ice is a) 80 cal/g b) 540 cal/g c) 100 cal/g d) 22.4 cal/g
82. Latent heat of vaporization of water is a) 80 cal/g b) 540 cal/g c) 100 cal/g d) 336 cal/g
83. RMS speed of gas molecules is proportional to a) $\sqrt{T}$ b) T c) $1/\sqrt{T}$ d) T^2
84. RMS speed of H_2 at STP is 2 km/s. RMS speed of O_2 at same temperature is a) 2 km/s b) 0.5 km/s c) 1 km/s d) 4 km/s

85. Average KE per degree of freedom per molecule is a) kT b) (1/2)kT c) (3/2)kT d) 2kT
86. Mean free path of gas molecules varies with pressure as a) p b) 1/p c) p^2 d) p^{-2}
87. Boyle's law states at constant temperature a) P∝V b) P∝1/V c) PV^2=const d) P/V=const
88. Charles's law states at constant pressure a) V∝T b) V∝1/T c) V∝P d) V=const
89. Ideal gas equation is a) PV=nRT b) PV=RT c) P=nRT/V (same as a) d) PT=nRV
90. A cooking gas cylinder withstands 20 atm. At 27°C, pressure is 10 atm. Cylinder bursts when temperature exceeds a) 273°C b) 327°C c) 300°C d) 127°C
91. First law of thermodynamics states a) ΔQ = ΔU + ΔW b) ΔU = ΔQ+ΔW c) ΔW=ΔQ+ΔU d) ΔQ=ΔU−ΔW
92. In an isothermal process, ΔU is a) positive b) negative c) zero d) infinite
93. In an adiabatic process, ΔQ is a) zero b) maximum c) equal to ΔU d) equal to ΔW only
94. Carnot engine efficiency between T_1 (source) and T_2 (sink) is a) $1-T_2/T_1$ b) $1-T_1/T_2$ c) T_2/T_1 d) T_1/T_2
95. A Carnot engine works between 600K and 300K. Efficiency is a) 25% b) 50% c) 75% d) 100%
96. Second law of thermodynamics implies a) heat flows spontaneously from cold to hot b) entropy of isolated system tends to increase c) energy is destroyed d) perpetual motion is possible
97. Stefan-Boltzmann law states radiated energy per unit area is proportional to a) T b) T^2 c) T^3 d) T^4
98. Two rods of same length/area, conductivities K_1, K_2 joined in series. Equivalent conductivity is a) $(K_1+K_2)/2$ b) $2K_1K_2/(K_1+K_2)$ c) K_1K_2 d) K_1+K_2
99. Heat transfer without a medium occurs by a) conduction b) convection c) radiation d) both a and b
100. Wien's displacement law relates temperature to a) total energy radiated b) wavelength of max emission c) frequency of light emitted d) speed of radiation

8. SHM, Waves & Sound (Q101–120)

101. Time period of SHM is T = a) $2\pi\sqrt{(k/m)}$ b) $2\pi\sqrt{(m/k)}$ c) $2\pi(m/k)$ d) $(1/2\pi)\sqrt{(m/k)}$
102. In SHM, velocity is maximum at a) extreme position b) mean position c) both equally d) nowhere
103. If time period of SHM is 12s, time for displacement to become half amplitude (from extreme) is a) 1s b) 2s c) 3s d) 4s
104. Simple pendulum's time period depends on a) mass of bob b) amplitude (small) c) length and g d) material of bob
105. If length of pendulum is doubled, time period becomes a) 2 times b) √2 times c) 4 times d) same
106. Relation v = fλ represents a) wave speed b) wave energy c) wave amplitude d) wave phase
107. Sound waves in air are a) transverse b) longitudinal c) both d) electromagnetic

108. Distance between two consecutive nodes in a stationary wave is a) λ b) $\lambda/2$ c) $\lambda/4$ d) 2λ

109. Speed of sound at 0°C is approximately a) 300 m/s b) 332 m/s c) 340 m/s d) 350 m/s

110. Speed of sound increases with increase in a) pressure at constant T b) temperature c) density d) wavelength

111. Doppler effect is due to a) change of medium b) relative motion between source and observer c) reflection d) diffraction

112. A source of frequency 150Hz moves toward observer at 110 m/s (v=330 m/s). Frequency heard is a) 150 Hz b) 200 Hz c) 225 Hz d) 100 Hz

113. Fundamental frequency of a closed organ pipe of length L is a) $v/2L$ b) $v/4L$ c) v/L d) $2v/L$

114. Fundamental frequency of an open organ pipe of length L is a) $v/2L$ b) $v/4L$ c) v/L d) $2v/L$

115. Open organ pipe produces a) only odd harmonics b) only even harmonics c) both odd and even harmonics d) no harmonics

116. Beats are produced due to a) interference of two waves of slightly different frequency b) reflection c) diffraction d) polarization

117. Number of beats per second equals a) sum of two frequencies b) difference of two frequencies c) product of frequencies d) average

118. Phase difference between two prongs of vibrating tuning fork is a) 0 b) $\pi/2$ c) π d) 2π

119. Open pipe and closed pipe resonate to same tuning fork. Ratio of lengths (open:closed) is a) 1:1 b) 2:1 c) 1:2 d) 4:1

120. Resonance occurs when frequency of applied force equals a) zero b) natural frequency of system c) any frequency d) infinite frequency

9. Ray Optics (Q121–140)

121. Focal length of a plane mirror is a) zero b) infinity c) equal to R d) negative

122. Mirror formula is a) $1/v+1/u=1/f$ b) $1/v-1/u=1/f$ c) $v+u=f$ d) $1/f=u/v$

123. Radius of curvature R and focal length f are related by a) $R=f$ b) $R=2f$ c) $R=f/2$ d) $R=4f$

124. Refractive index is defined as a) speed in medium/speed in vacuum b) speed in vacuum/speed in medium c) wavelength ratio only d) frequency ratio

125. Snell's law states a) $n_1\sin\theta_1=n_2\sin\theta_2$ b) $n_1\cos\theta_1=n_2\cos\theta_2$ c) $n_1/\sin\theta_1=n_2/\sin\theta_2$ d) $n_1\theta_1=n_2\theta_2$

126. Critical angle is the angle of incidence for which angle of refraction is a) 0° b) 45° c) 90° d) 180°

127. If critical angle for a medium is 30°, refractive index is a) 0.5 b) 1.5 c) 2 d) 3

128. Lens maker's formula is a) $1/f=(n-1)(1/R_1-1/R_2)$ b) $1/f=n(1/R_1+1/R_2)$ c) $f=(n-1)(R_1+R_2)$ d) $1/f=(n+1)(1/R_1-1/R_2)$

129. Power of a lens is a) f b) $1/f$ (in meters) c) f^2 d) $1/f^2$

130. Two thin lenses +5D and −7.5D in contact. Combined power is a) 12.5D b) −2.5D c) 2.5D d) −12.5D

131. When a convex lens is immersed in water, focal length a) decreases b) increases c) remains same d) becomes negative

132. Dispersion of light through a prism is due to a) different refractive index for different wavelengths b) reflection c) diffraction d) polarization

133. Which color deviates most through a prism? a) Red b) Violet c) Yellow d) Green

134. A myopic eye is corrected using a a) convex lens b) concave lens c) cylindrical lens d) bifocal lens

135. A hypermetropic eye is corrected using a a) convex lens b) concave lens c) plane lens d) prism

136. Near point of a normal eye is at a) infinity b) 25 cm c) 50 cm d) 2.5 cm

137. Magnifying power of a simple microscope (near point D=25cm, focal length f) is a) D/f b) 1+D/f c) f/D d) D×f

138. In an astronomical telescope, magnifying power (normal adjustment) is a) f_0/f_e b) f_e/f_0 c) $f_0 \times f_e$ d) f_0+f_e

139. Young's double slit experiment demonstrates a) rectilinear propagation b) interference of light c) polarization d) dispersion

140. Fringe width in YDSE is β = a) λD/d b) λd/D c) Dd/λ d) λ/Dd

10. Electrostatics (Q141–155)

141. Coulomb's law is analogous to a) Newton's law of gravitation b) Ohm's law c) Faraday's law d) Ampere's law

142. If distance between two charges is doubled, force becomes a) doubled b) halved c) one-fourth d) four times

143. Electric field due to a point charge varies with distance as a) 1/r b) $1/r^2$ c) r^2 d) r

144. Electric potential due to point charge Q at distance r is a) kQ/r b) kQ/r^2 c) kQr d) kQ^2/r

145. Relation between electric field E and potential V is a) E=−dV/dx b) E=dV/dx c) $E=V/x^2$ d) V=Ex always

146. Electric field inside a conductor in electrostatic equilibrium is a) maximum b) zero c) infinite d) equal to surface field

147. Electric field at surface of a charged conductor is a) tangential to surface b) normal to surface c) at 45° d) zero

148. Dipole moment is given by a) p=qd b) p=q/d c) $p=q^2d$ d) $p=qd^2$

149. Torque on electric dipole in uniform field E is a) τ=pE b) τ=pE sinθ c) τ=pE cosθ d) τ=p/E

150. Capacitance of a parallel plate capacitor is a) $\varepsilon_0 A/d$ b) $\varepsilon_0 d/A$ c) $A/\varepsilon_0 d$ d) $\varepsilon_0 Ad$

151. Three capacitors 3μF, 9μF, 18μF in series. Equivalent capacitance is a) 30μF b) 2μF c) 1.8μF d) 15μF

152. Energy stored in a capacitor is a) QV b) (1/2)QV c) Q/V d) 2QV

153. Inserting a dielectric between capacitor plates a) decreases capacitance b) increases capacitance c) no effect d) makes capacitance infinite

154. A capacitor blocks a) AC only b) DC (steady state) c) both equally d) neither

155. Electric field due to an infinite sheet of charge is a) dependent on distance b) independent of distance c) zero d) inversely proportional to r^2

11. Current Electricity (Q156–170)

156. Ohm's law states V = a) IR b) I/R c) I^2R d) I+R

157. Resistance of a wire depends on a) length, area, material (resistivity) b) current only c) voltage only d) temperature only

158. Two resistors 4Ω, 6Ω in series. Equivalent resistance is a) 2.4Ω b) 10Ω c) 24Ω d) 1.5Ω

159. Two resistors 4Ω, 6Ω in parallel. Equivalent resistance is a) 2.4Ω b) 10Ω c) 24Ω d) 1.5Ω

160. If length of a wire doubled (constant volume), resistance becomes a) 2 times b) 4 times c) half d) same

161. Kirchhoff's current law is based on conservation of a) energy b) charge c) momentum d) mass

162. Kirchhoff's voltage law is based on conservation of a) charge b) energy c) momentum d) mass

163. A cell of EMF X connected to resistance R gives terminal PD Y. Internal resistance is a) (X−Y)R/Y b) (X−Y)/R c) XR/Y d) (X+Y)R/Y

164. Wheatstone bridge is used to measure a) EMF b) current c) unknown resistance d) capacitance

165. Heat produced in a resistor is (Joule's law) a) $H=I^2Rt$ b) H=IRt c) H=IR/t d) $H=I^2R/t$

166. Resistance of an ideal ammeter is a) infinite b) zero c) very high d) equal to circuit resistance

167. A galvanometer is converted into an ammeter by connecting a) high resistance in series b) low resistance (shunt) in parallel c) high resistance in parallel d) low resistance in series

168. Fuse wire should have a) high resistivity, high melting point b) low resistivity, low melting point c) high resistivity, low melting point d) low resistivity, high melting point

169. Drift velocity of electrons in a conductor is a) very high (near light speed) b) very low (mm/s range) c) equal to speed of light d) zero

170. 1 unit of electricity equals a) 1 kWh b) 1 kW c) 1 Wh d) 1000 J

12. Magnetism & EMI (Q171–182)

171. Magnetic field inside a long solenoid is given by a) $B=\mu_0 nI$ b) $B=\mu_0 n/I$ c) $B=\mu_0 I/n$ d) $B=\mu_0/nI$

172. Force on a current-carrying conductor in magnetic field is a) F=BIL b) F=BIL sinθ c) F=BI/L d) F=B/IL

173. Two parallel wires carrying current in same direction a) repel b) attract c) no force d) rotate

174. Radius of circular path of charged particle in magnetic field is a) r=mv/qB b) r=qB/mv c) r=mB/qv d) r=qv/mB

175. Angle of dip is the angle between a) magnetic and geographic meridian b) Earth's total field and horizontal direction c) vertical and horizontal only d) two poles

176. At the magnetic equator, angle of dip is a) 0° b) 90° c) 45° d) 180°

177. Electromagnetic induction was discovered by a) Ampere b) Faraday c) Oersted d) Maxwell

178. Lenz's law is a consequence of conservation of a) charge b) energy c) momentum d) mass

179. Induced EMF in coil of N turns is a) EMF=N(dΦ/dt) b) EMF=−N(dΦ/dt) c) EMF=Φ/N d) EMF=NΦt

180. Self-inductance of coil is 10mH, current 2A flows. Magnetic flux through coil is a) 0.02 Wb b) 0.01 Wb c) 0.05 Wb d) 20 Wb

181. A transformer works on the principle of a) self-induction b) mutual induction c) electrostatic induction d) Coulomb's law

182. For a step-up transformer, secondary turns compared to primary are a) less b) more c) equal d) zero

13. AC Circuits (Q183–188)

183. RMS value of AC current $I=I_0\sin(\omega t)$ is a) I_0 b) $I_0/\sqrt{2}$ c) $I_0\sqrt{2}$ d) $I_0/2$

184. Frequency of AC supply in Nepal is a) 60 Hz b) 50 Hz c) 100 Hz d) 25 Hz

185. Impedance in series LCR circuit is a) Z=R+XL+XC b) $Z=\sqrt{(R^2+(XL-XC)^2)}$ c) $Z=R^2+XL^2+XC^2$ d) Z=R/(XL+XC)

186. Resonance in a series LCR circuit occurs when a) XL=0 b) XC=0 c) XL=XC d) R=0

187. At resonance, impedance of series LCR circuit is a) maximum b) minimum, equal to R c) zero d) infinite

188. Quality factor Q of series resonant circuit is a) XL/R (at resonance) b) R/XL c) XL×R d) 1/(XLR)

14. Modern Physics — Photoelectric, Bohr Model, X-rays (Q189–200)

189. Einstein's photoelectric equation is a) hν=W+KE b) hν=W−KE c) hν=KE−W d) W=hν×KE

190. Stopping potential depends on a) intensity of light b) frequency of light and work function c) area of metal d) distance of source

191. de Broglie wavelength is given by a) $\lambda=h/p$ b) $\lambda=hp$ c) $\lambda=p/h$ d) $\lambda=h/mc$

192. If kinetic energy of a particle is doubled, de Broglie wavelength becomes a) 2 times b) 4 times c) $1/\sqrt{2}$ times d) $\sqrt{2}$ times

193. Radius of nth Bohr orbit is proportional to a) n b) n^2 c) 1/n d) $1/n^2$

194. Energy of hydrogen atom in ground state is a) 0 eV b) −13.6 eV c) +13.6 eV d) −3.4 eV

195. The Balmer series lies in which region? a) Ultraviolet b) Visible c) Infrared d) X-ray

196. Half-life of a radioactive element is T. Mean life (τ) relation is a) $\tau=T$ b) $\tau=T/0.693$ c) $\tau=0.693T$ d) $\tau=T^2$

197. A radioactive material of half-life 5 days, initial mass 1kg. Amount remaining after 20 days is a) 125g b) 250g c) 62.5g d) 500g

198. Binding energy of a nucleus relates to mass defect Δm by a) $E=\Delta mc$ b) $E=\Delta mc^2$ c) $E=\Delta m/c^2$ d) $E=\Delta m^2c$

199. NPN transistors are preferred over PNP mainly because a) they are cheaper b) electrons have higher mobility than holes c) they consume more power d) easier to manufacture

200. NAND and NOR gates are called universal gates because a) they are the fastest gates b) any logic gate can be built using only NAND (or only NOR) c) they consume least power d) invented first

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Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	a	41	a	81	a	121	a	161	b
2	a	42	b	82	b	122	a	162	b
3	b	43	a	83	a	123	b	163	a
4	b	44	b	84	c	124	b	164	c
5	b	45	b	85	b	125	a	165	a
6	a	46	c	86	b	126	c	166	b
7	a	47	b	87	b	127	c	167	b
8	a	48	b	88	a	128	a	168	c
9	a	49	b	89	a	129	b	169	b
10	b	50	a	90	b	130	b	170	a
11	c	51	b	91	a	131	b	171	a
12	a	52	b	92	c	132	a	172	b

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
13	b	53	b	93	a	133	b	173	b
14	b	54	b	94	a	134	b	174	a
15	b	55	b	95	b	135	a	175	b
16	a	56	b	96	b	136	b	176	a
17	b	57	b	97	d	137	b	177	b
18	a	58	b	98	b	138	a	178	b
19	a	59	c	99	c	139	b	179	b
20	a	60	b	100	b	140	a	180	a
21	a	61	b	101	b	141	a	181	b
22	a	62	c	102	b	142	c	182	b
23	b	63	c	103	c	143	b	183	b
24	a	64	a	104	c	144	a	184	b
25	a	65	b	105	b	145	a	185	b
26	c	66	b	106	a	146	b	186	c
27	b	67	b	107	b	147	b	187	b
28	a	68	a	108	b	148	a	188	a
29	a	69	b	109	b	149	b	189	a
30	a	70	b	110	b	150	a	190	b
31	a	71	b	111	b	151	b	191	a
32	b	72	b	112	c	152	b	192	c
33	b	73	a	113	b	153	b	193	b
34	b	74	c	114	a	154	b	194	b
35	a	75	a	115	c	155	b	195	b
36	a	76	c	116	a	156	a	196	b
37	a	77	b	117	b	157	a	197	c
38	b	78	b	118	c	158	b	198	b
39	b	79	a	119	b	159	a	199	b

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
40	b	80	b	120	b	160	b	200	b

Priority note: This 200-question set is your **core revision spine** for Physics. If time is limited before the exam, master this set first — it concentrates the highest-frequency formulas and concepts (Heat/Thermo, Current Electricity, Ray Optics, Electrostatics, Rotational Mechanics, Bohr Model/Radioactivity) that have appeared with the greatest consistency across 2069–2081 official papers.